

Section 11.2 Vectors in Space

(1) **Definition:** A vector in $\mathbb{R}^3$ is an ordered triple of real numbers

example: $(2, 3, 0)$ which is $x = 2$, $y = 3$, and $z = 0$

We have 3 options to express vectors

Option 1: $\langle 2, 3, 0 \rangle$

Option 2: $\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$

Option 3: $2\mathbf{i} + 3\mathbf{j} + 0\mathbf{k}$ (I means this is vertically listed)

Special Option 4 will let you express vector based on the x , y , and z components, but it's not something usual, so I'm not going to talk about it.

Some Operations on vectors

(a) Vectors may be added or subtracted by $\pm$ components.

(b) A constant may be multiplied by vector by multiplied each components.

example: $(2, 2, 4) + 3(0, 0, 0) = (2, 2, 4) + (0, 0, 0) = (2, 2, 4)$

Another definition:

$\mathbf{i}$ hat = $(1, 0, 0)$

$\mathbf{j}$ hat = $(0, 1, 0)$

$\mathbf{k}$ hat = $(0, 0, 1)$

from the example above, it could be represent as:

$(2, 2, 4) = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

Some additional notes:

- In $\mathbb{R}^3$, there is no $\mathbf{i}$ hat
- Don't forget to write coefficient 1, example: $1\mathbf{i}$ hat = $\mathbf{i}$ hat, $0\mathbf{k}$ hat could just be represent as $\mathbf{i}$ hat = $\mathbf{i}$ hat + $0\mathbf{k}$ hat
- To denote a vector, we always add a line above it's name.

example:

$\mathbf{v}$ or $\vec{v}$

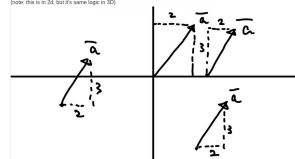
(2) Representation of Vectors Visually

A vector can be represented by arrow which points in direction specified by it's components, where it's anchored doesn't matter.

example:

$\mathbf{v} = 2\mathbf{i}$ hat + $3\mathbf{j}$ hat

from this is in $\mathbb{R}^3$, but it's same logic in $\mathbb{R}^2$



(3) **Norm:** often we'll anchor vectors at specific locations for specific reasons.

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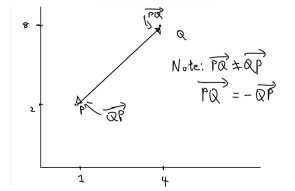
(100) **Norm:** often we'll anchor vectors at specific locations for specific reasons.

(4) The zero vector is denoted by $\vec{0}$, obviously, everything of there is 0 for $\mathbf{i}$ hat, $\mathbf{j}$ hat, $\mathbf{k}$ hat.

(5) Given 2 points $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$. The vector pointing from P to Q is $\vec{PQ}$ and it's $\vec{PQ} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$.

example: $P = (1, 2, 3)$ and $Q = (4, 5, 6)$

$\vec{PQ} = (4 - 1)\mathbf{i} + (5 - 2)\mathbf{j} + (6 - 3)\mathbf{k} = 3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$



Note: $\vec{PQ} \neq \vec{QP}$

$\vec{PQ} = -\vec{QP}$

(6) Two vectors $\vec{u}$ and $\vec{v}$ are parallel if they are non-zero scalar multiples of another.

example: are parallel case:

$\vec{u} = 4\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$ and $\vec{v} = 20\mathbf{i} + 30\mathbf{j} + 40\mathbf{k}$

$\vec{v} = 5\vec{u}$ so they are parallel.

example: are not parallel case:

$\vec{u} = 4\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$ and $\vec{v} = 10\mathbf{i} + 15\mathbf{j} + 20\mathbf{k}$

$\vec{v} = 2.5\vec{u}$ so they are not parallel.

(7) **Line of Motion**

(a) To represent location: vector vector at the origin, then x, y, z are the location from the location of $\vec{0}$.

(b) To represent velocity: vector vector at the origin, then x, y, z are the velocity of that objects of each direction.

(c) To represent force: vector vector at the origin, adding vector which represent forces gives resultant forces.

for example: an object at the origin is acted on by two forces.

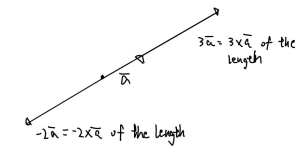
$\vec{F}_1 = 4\mathbf{i} + 6\mathbf{j} + 8\mathbf{k}$ and $\vec{F}_2 = 10\mathbf{i} + 15\mathbf{j} + 20\mathbf{k}$

$\vec{F} = \vec{F}_1 + \vec{F}_2 = 14\mathbf{i} + 21\mathbf{j} + 28\mathbf{k}$

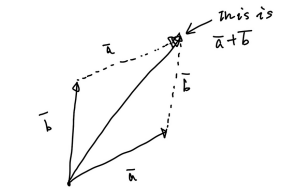
The result then here: $\vec{F} = 14\mathbf{i} + 21\mathbf{j} + 28\mathbf{k}$

(8) **Unit:** Unitless, Representation of vector operations:

(a) **Scalar multiplication:**



(b) **Vector addition:**



(c) **Vector subtraction:**

